

Question Paper

Exam Date & Time: 26-Dec-2025 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU
FIFTH SEMESTER B.TECH. DEGREE MAKEUP EXAMINATIONS - DECEMBER 2025

Principles of Computer Vision [CSE_3125]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Interpret the image processing workflow and explain with an example how a raw satellite image can be transformed into a usable form for analysis. (5)

A)

Course Outcome: "CO1" / Cognitive Level: "Understand"

- B) Let p and q be the image pixels at coordinates (150,150) and (75,25) respectively. Compute the distance between these pixels using different distance measures and identify which distance measure gives the minimum distance between the given pixels. (5)

Consider 3-bit image of size 8X8 shown below. Calculate histogram of the given image and apply histogram equalization. Show the output histogram.

3	4	4	5	6	6	5	4
2	3	4	5	6	7	6	5
1	2	3	4	5	6	6	5
0	1	2	3	4	5	6	7
0	0	1	2	3	4	5	6
1	2	3	4	5	6	7	7
2	3	4	5	6	6	5	4
3	4	5	6	7	7	6	5

Course Outcome: "CO1" / Cognitive Level: "Application"

- 2) Illustrate discontinuity in the context of image processing. (5)

List the various types of discontinuities and explain how each type can be detected using appropriate masks or operators.

A)

Course Outcome: "CO2" / Cognitive Level: "Understand"

- B) (5)

The Mean Shift Algorithm (MSS) is a non-parametric clustering technique that shifts data points toward regions of higher data density until convergence.

Solve using MSS with a bandwidth $h = 3$ to find the cluster centers for the following set of 2D data points:

$(1, 2), (2, 3), (3, 3), (6, 6), (7, 7), (8, 7)$

Course Outcome: "CO2" / Cognitive Level: "Application"

- 3) A satellite captures images of Earth's surface at different times, but because of orbital tilt and Earth's curvature, the images appear distorted or misaligned. Explain how geometric transformations help in correcting distortions and aligning multiple satellite images for accurate mapping. (5)
- A)

Course Outcome: "CO3" / Cognitive Level: "Understand"

- B) A robotics startup develops a self-driving car that uses two calibrated cameras to estimate distances to nearby objects. The system compares two images to find corresponding points and calculate depth. Apply stereo correspondence and triangulation to explain how a calibrated stereo vision system can estimate the 3D position of objects from 2D image pairs. (5)

Course Outcome: "CO3" / Cognitive Level: "Application"

- 4) A sports analysis system captures continuous video of players during a match. The software tracks their motion to generate speed statistics. Apply motion estimation and hierarchical search techniques to detect and refine player movement across consecutive frames. (5)
- A)

Course Outcome: "CO4" / Cognitive Level: "Application"

- B) A video sequence is corrupted by additive Gaussian noise. (5)
- (i) Analyze why multi-frame, motion-compensated video denoising produces better visual and statistical results than applying a spatial filter independently to each frame.
- (ii) Justify, with reasoning, whether multi-frame denoising always performs better than single-frame denoising for high-motion and low-frame-rate videos.

Course Outcome: "CO4" / Cognitive Level: "Analysis"

- 5) Apply an object detection and classification pipeline to develop a leaf disease detection system for farmers using mobile phone images. Give details of the methods you apply to segment the leaf region, use feature extraction techniques, and implement a trained model to classify the disease type. (5)
- A)

Course Outcome: "CO5" / Cognitive Level: "Application"

- B) A smart assisted-living system is being deployed in the homes of elderly individuals to monitor daily activities such as walking, sitting, sleeping, and falling. The system uses a fixed indoor camera and must work without wearable sensors. It should detect abnormal situations such as a fall and send alerts to caregivers. Apply a computer-vision-based activity detection model (human detection, extract pose features and apply a trained activity recognition model) to implement this elderly monitoring system. Explain how you would apply these techniques to automatically identify normal vs. abnormal activities and trigger alerts in real-time. (5)

Course Outcome: "CO5" / Cognitive Level: "Application"

-----End-----

Event Name	Subject
------------	---------

058-MIT BLR-MAKEUP EXAM-2025-2026

CSE_3125:Principles of Computer Vision

#	Course Outcome	Description	Marks
1	1	CO 1	10
2	2	CO 2	10
3	3	CO 3	10
4	4	CO 4	10
5	5	CO 5	10
Total			50

* **NA** - CO Not Selected

Event Name	Subject
------------	---------

058-MIT BLR-MAKEUP EXAM-2025-2026

CSE_3125:Principles of Computer Vision

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	3	15(30%)
3	Application	6	30(60%)
4	Analysis	1	5(10%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
Total		10	50 (100%)